Limits at infinity

In this session[13] we are going to see how to compute the limit of a function when the domain goes to $\pm\infty$.

[13] January 29

Properties of Limits

Before delving into computing limits of rational functions, it is important to mention that we are going to need some properties of limits[14]. The demostrations is a little bit outside the scope of this course, but if you are curious in off, you can ask me.

[14]

$\lim_{x\to c} b = b$	Limit of a constant
$\lim_{x\to c} x = c$	Direct substitution technique
$\lim_{x\to c} x^n = c^n$	Direct substitution

Assumming that b and c are real numbers, n a positive integer and let $f(x)$ and $g(x)$ be functions with the following limits $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} f(x) = K$.

Scalar multiple:	$\lim_{x\to c}[bf(x)] = bL$
Sum or Diff:	$\lim_{x\to c}[f(x)\pm g(x)] = L \pm K$
Product:	$\lim_{x\to c}[f(x)g(x)] = LK$
Quotient:	$\lim_{x\to c}\left[\frac{f(x)}{g(x)}\right] = \frac{L}{K}$
Power:	$\lim_{x\to c}[f(x)]^n = L^n$

Rational functions

Let's review a helpful limit to examine any rational function before we begin learning how to compute these limits $\lim_{\pm\infty} \frac{1}{x} = 0$. This limit can be easily proved using a numerical approach.

Now, let's remember that a rational function is expressed as,

$$f(x) = \frac{P(x)}{Q(x)}.$$

With $P(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_0$ and $Q(x) = b_m x^m + b_{m-1}x^{m-1} + \ldots + b_0$. In order to compute the limit to $\pm\infty$, first factorize an x. Then, simplify the expression algebraically and apply the previously established properties of limits. For example to compute $\lim_{x\to\infty} \frac{x+2}{x^2+1}$ we need to rewrite[15] the function as

$$\lim_{x\to\infty} \frac{x+2}{x^2+1} = \lim_{x\to\infty} \frac{1+2/x}{x+1/x}.$$

[15]
$$\begin{aligned}\frac{x+2}{x+1} &= \frac{x(1+2/x)}{x(x+1/x)}\\ &= \frac{1+2/x}{x+1/x}\end{aligned}$$

Applying the properties of limits[16],

$$\lim_{x\to\infty} \frac{x+2}{x^2+1} = \frac{1}{\infty} = 0.$$

[16]
$$\begin{aligned}\lim_{x\to\infty} \frac{1+2/x}{x+1/x} &= \frac{\lim_{x\to\infty} 1 + \lim_{x\to\infty} 2/x}{\lim_{x\to\infty} x + \lim_{x\to\infty} 1/x}\\ &= \frac{1+0}{\infty+0}\end{aligned}$$

Another approach can be by dividing by the highest factor[17], in this case will be x^2, therefore:

$$\lim_{x\to\infty} \frac{x+2}{x^2+1} = \lim_{x\to\infty} \frac{1/x+2/x^2}{1+1/x^2}.$$

[17]
$$\begin{aligned}\frac{x+2}{x^2+1} &= \frac{x/x^2+2/x^2}{x^2/x^2+1/x^2}\\ &= \frac{1/x+2/x^2}{1+1/x^2}\end{aligned}$$

Again, applying properties of limits[18],

$$\lim_{x\to\infty} \frac{x+2}{x^2+1} = \frac{0}{1} = 0.$$

[18]
$$\begin{aligned}\lim_{x\to\infty} \frac{1/x+2/x^2}{1+1/x^2} &= \frac{\lim_{x\to\infty} 1/x + \lim_{x\to\infty} 2/x^2}{\lim_{x\to\infty} 1 + \lim_{x\to\infty} 1/x^2}\\ &= \frac{0+0}{1+0}\\ &= \frac{0}{1}\end{aligned}$$

Class Activities

$$\lim_{x\to\infty} \frac{3x+4}{2x^2+5}, \quad \lim_{x\to\infty} \frac{3x^2+4}{2x+5}, \quad \lim_{x\to\infty} \frac{3x^2+4}{2x^2+5}$$